An Explanation for Molecular Déjà Vu

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Abstract

The rapid migration of large language models into chemistry, materials science and molecular-property prediction has created an unusual epistemological difficulty: when an artificial-intelligence system supplies the correct numerical property of a molecule, has it actually predicted that property, or has it encountered and retained the published value during training? This distinction lies at the heart of the newly identified phenomenon aptly termed **"Molecular Déjà Vu."** Recent work by Busch *et al.* demonstrates that digit-level retrieval of published molecular properties occurs across a significant subset of contemporary frontier language models and established molecular regression benchmarks [1].

Intriguingly, increasing a model's reasoning effort can increase rather than suppress apparent retrieval. This article develops an equation less conceptual explanation of molecular déjà vu, distinguishes memorisation from chemical inference, examines the peculiar role played by molecular representations such as SMILES, considers benchmark contamination and representation dependence, and explores the implications for AI-assisted chemistry. The central contention is that molecular déjà vu is neither straightforward fraud nor proof that language models cannot reason chemically. Rather, it exposes an increasingly porous boundary between **knowledge, memory, recognition and prediction**. Reliable molecular AI will consequently require evaluation protocols specifically designed to determine not merely whether a model knows the right answer, but whether it can reach that answer when remembering it is impossible.

Keywords: *Molecular déjà vu; large language models; molecular property prediction; benchmark contamination; memorisation; SMILES; molecular representation; frontier models; chemical artificial intelligence; digit-level retrieval; MoleculeNet; scientific reasoning; generalisation; in-context learning.*

1. Introduction: When Prediction Looks Suspiciously Like Memory

Déjà vu is ordinarily a peculiarity of human consciousness: the eerie sensation that an unfamiliar moment has somehow happened before.

Artificial intelligence now appears capable of something metaphorically similar.

Present a large language model with the representation of a molecule and ask for its experimentally determined solubility, lipophilicity, atomisation energy or another numerical property. The model may return an impressively accurate number.

At first sight this appears to constitute **molecular-property prediction**.

But suppose the number is not merely close.

Suppose it reproduces the published benchmark value **digit for digit**.

Has the machine inferred a chemical property—or recognised an old acquaintance?

That is the problem raised forcefully by Busch *et al.* in their 2026 study *Molecular Déjà Vu: Digit-Level Retrieval of Published Values in Frontier Language Models* [1]. The investigators audited **22 frontier language models across 12 molecular regression benchmarks** and found that verbatim numerical retrieval was not a rare curiosity. On five datasets, more than half of the tested models exhibited evidence of it; on other datasets the phenomenon was much less frequent.

The distinction matters enormously because molecular-property prediction is becoming an important application of AI in drug discovery, materials design and computational chemistry [2,3].

A model that can infer an unknown property may assist scientific discovery.

A model that remembers a known property may merely possess an excellent library.

Confusing the latter with the former creates **an illusion of scientific intelligence**.

2. What Exactly Is Molecular Déjà Vu?

“Molecular déjà vu” should not be understood as a molecule itself possessing memory. Rather, it describes an AI model apparently **recognising molecular information encountered somewhere in its enormous pretraining history**.

Consider a publicly available chemical database containing a molecule represented by its SMILES string together with an experimentally measured property.

That database may subsequently appear in repositories, papers, supplementary information, educational websites, software documentation and machine-learning benchmarks. Large language models are trained upon immense collections assembled from precisely such digital material.

The model can therefore encounter some combination of:

molecular representation → property name → numerical value.

Years later, an evaluator presents the same molecule and asks the model for the property.

The model returns the correct number.

Ordinary benchmarking records a triumph.

Yet nothing in that observation alone establishes that the model performed a chemical prediction.

The model may simply have encountered sufficiently similar information during training.

This is molecular déjà vu: **the benchmark asks the model to discover something that, unbeknown to the evaluator, the model may already remember.**

3. Why Digit-Level Retrieval Is So Revealing

There is an important difference between approximately predicting a molecular quantity and reproducing its published decimal expansion.

A scientifically useful predictor should generally possess some uncertainty. Molecular properties depend upon structure, experimental conditions, measurement conventions and modelling assumptions.

If a model estimates a property close to the measured value, genuine inference is plausible.

But an exact string of digits is different.

The probability that a general-purpose language model independently reconstructs an arbitrary published experimental decimal with exact precision can become suspiciously small.

Digits therefore behave rather like **molecular fingerprints of memory**.

Busch and colleagues exploit precisely this distinction. Rather than asking only how small the prediction error is, they investigate whether model outputs contain evidence that specific published values are being retrieved [1].

That transforms the evaluation question from:

“How accurate is the model?”

into:

“Why is the model accurate?”

This is the deeper achievement of the work.

4. The Benchmark Contamination Problem

The problem extends far beyond chemistry.

Modern LLM evaluation suffers from what is generally called **benchmark contamination**: material intended to test a model may already have occurred in its training corpus.

Deng *et al.* demonstrated that established language benchmarks can exhibit striking signs of such contamination, while Dong *et al.* developed distribution-based methods for distinguishing memorisation from generalisation [4,5]. Golchin and Surdeanu subsequently introduced the Data Contamination Quiz, designed specifically to detect whether models recognise original benchmark instances among carefully altered alternatives [6].

Chemistry presents an especially interesting version of this problem.

The celebrated **MoleculeNet** collection was deliberately created as a public benchmark suite for molecular machine learning [7]. Its accessibility is scientifically valuable: researchers can reproduce results and compare algorithms.

But publicity has an unintended consequence in the LLM era.

The longer a benchmark remains publicly available, the more opportunities exist for its molecules, labels and associated publications to migrate into gigantic training corpora.

The benchmark eventually becomes famous enough that the examination paper may inadvertently find its way into the student's library.

5. SMILES: Chemistry Becomes Language

Another reason molecular déjà vu is possible lies in the unusual relationship between chemistry and text.

A molecule is a physical object, but modern cheminformatics routinely represents molecules using strings such as **SMILES**.

To a chemist, a SMILES string encodes molecular connectivity.

To an LLM, however, it is also a sequence of tokens.

That dual identity is crucial.

An LLM need not possess a three-dimensional internal picture of a molecule to recognise that it has encountered a particular character sequence—or a related sequence—before.

Indeed, recent research reveals considerable sensitivity to molecular representation. A 2025 EMNLP study comparing SMILES, Deep SMILES, SELFIES, InChI and IUPAC representations found substantial differences in LLM molecular-property performance depending upon how the same chemistry was expressed [8].

Even more strikingly, another 2025 investigation found extremely poor consistency when state-of-the-art general-purpose LLMs were given equivalent molecules in different representational forms. Reported consistency across SMILES and IUPAC formulations could be **1% or less** in the investigated settings [9].

That finding should make us cautious.

A system possessing a genuinely representation-independent chemical concept ought ideally to recognise that two faithful descriptions correspond to the same underlying molecule.

If changing the spelling of the chemistry changes the answer dramatically, some portion of apparent chemical intelligence may reside in **linguistic pattern recognition rather than molecular understanding**.

6. The Extraordinary Reasoning Paradox

The most fascinating result of the new molecular-déjà-vu study concerns reasoning.

One might naturally predict that increasing reasoning effort would force an LLM away from memorisation and toward genuine inference.

Busch *et al.* found something considerably stranger.

Using the same molecules and prompts, experiments conducted at higher reasoning levels were flagged for retrieval **89% more often** than those at the lowest reasoning level [1].

This does not mean reasoning is intrinsically harmful.

It suggests instead that additional computational deliberation may give a sufficiently capable model more opportunity to

search or reconstruct information already encoded within its parameters.

Reasoning and remembering are therefore not necessarily adversaries.

Reasoning may become a better excavator of memory.

This is an important conceptual correction.

A lengthy chain of reasoning is not itself proof that the final answer was independently derived. The reasoning process may rationalise, reconstruct or contextualise a destination that memory has already supplied.

A student who remembers that the answer is 7.43 can still produce an elaborate derivation ending at 7.43.

The derivation does not establish that memory played no role.

7. Can the Memory Be Broken?

Here the investigators performed a particularly clever experiment.

They attempted to interfere with retrieval by transforming molecular representations while retaining the underlying chemistry. In strongly contaminated cases, however, some of the most capable models could still recognise combinations involving transformed SMILES strings and the original labels [1].

That tells us something profound about modern LLM memory.

Memorisation need not resemble a photographic archive.

A model may not simply store:

string X = number Y.

Instead, training can establish a distributed association among molecular names, textual representations, benchmark contexts, property labels and numerical values.

Consequently, superficial disguise may fail.

The model experiences something closer to genuine déjà vu: the presentation has changed, yet enough latent familiarity survives for recognition.

8. Prediction, Retrieval and the Grey Territory Between Them

It would nevertheless be a mistake to conclude that LLM molecular prediction is *nothing but memorisation*.

The 2026 results themselves argue against that simplistic interpretation.

When retrieval was suppressed, differences among models' prediction errors became smaller, but they did not disappear. The authors therefore conclude that general predictive capability cannot be explained solely by the quantity of memorised values [1].

This produces three conceptually different regimes.

At one extreme lies **pure retrieval**: the molecule and its property are remembered.

At the opposite extreme lies **genuine extrapolation**: the model encounters a genuinely novel molecular situation and derives a useful estimate from learned chemical relationships.

Between them lies the enormous and scientifically troublesome territory of **memory-assisted inference**.

A model may remember fragments: similar molecules, homologous compounds, approximate ranges, chemical rules, published trends or neighbouring examples.

But human scientists do precisely this.

A chemist asked to estimate the boiling point of an unfamiliar compound does not erase decades of remembered chemistry before answering. Scientific reasoning itself consists partly of applying accumulated memory to novel situations.

Thus the real question is not whether an AI system uses memory.

Of course it does.

The question is whether the evaluation measures its capacity to **generalise beyond memory**.

9. Molecular Déjà Vu Is an Epistemological Problem

This moves the issue beyond computer science.

Suppose an AI correctly predicts a previously measured molecular property because the number exists somewhere within its training history.

Scientifically, no discovery has occurred.

Suppose instead that the AI predicts an unmeasured property and experiment later confirms it.

That constitutes evidence of genuine predictive utility.

The observable output may look identical in both cases.

The difference lies entirely in the **provenance of knowledge**.

Molecular déjà vu therefore reveals an epistemological weakness in contemporary AI benchmarking: correctness alone cannot tell us whether a machine **knew, remembered, inferred, interpolated or discovered**.

The distinction becomes particularly serious when claims concern scientific reasoning.

10. Toward Déjà-Vu-Resistant Molecular AI

Future molecular benchmarks should consequently become more adversarial.

First, evaluation molecules should ideally **post-date the model's training cutoff**.

Second, benchmark designers can use newly synthesised compounds or newly measured properties whose answers could not have appeared in pretraining.

Third, chemically equivalent but textually altered representations should be employed to test representation invariance.

Fourth, evaluations should deliberately include **out-of-distribution molecular scaffolds**, rather than near neighbours of famous benchmark compounds.

Fifth, experimental laboratories can create genuinely blind prospective challenges: provide molecular structures today, record AI predictions, perform experiments tomorrow, and reveal the measurements only afterward.

That is the cleanest antidote to déjà vu.

The machine cannot remember tomorrow's experiment.

Reasoning-driven benchmark construction may also help. Recent work suggests that evaluation requiring newly synthesised multi-step reasoning can be more resistant to shallow benchmark memorisation than direct retrieval-style questions [10].

11. Research Findings

The emerging literature supports several principal findings.

First, digit-level retrieval of molecular benchmark values appears sufficiently widespread among frontier LLMs that conventional accuracy scores cannot automatically be interpreted as evidence of molecular predictive ability [1].

Second, contamination is strongly benchmark-dependent. Some molecular datasets appear substantially more susceptible than others, indicating that exposure history matters.

Third, additional reasoning effort does not necessarily eliminate memorisation and may actually facilitate retrieval.

Fourth, molecular representation matters enormously. Equivalent chemistry expressed through SMILES, IUPAC,

InChI or alternative strings can generate materially different behaviour [8,9].

Fifth, suppressing identifiable retrieval does not eliminate differences in model performance. There remains evidence for genuine predictive capacity beyond straightforward memorisation [1].

Finally, the most trustworthy test of molecular intelligence will increasingly be prospective rather than retrospective: prediction of chemical measurements that literally **did not exist when the model was trained**.

12. Conclusion: The Molecule the Machine Thinks It Has Met Before

Molecular déjà vu exposes one of the strangest consequences of marrying gigantic language models to the accumulated literature of science.

Humanity has spent centuries publishing measurements.

We have now trained machines upon enormous portions of that published inheritance and begun asking those machines to predict the very measurements we previously published.

When they answer correctly, we applaud their powers of prediction.

Sometimes, perhaps, **they simply recognise the question**.

Yet this should not be regarded merely as an embarrassment. Molecular déjà vu provides an extraordinarily useful diagnostic window into how scientific knowledge resides within frontier AI systems. These models appear capable of blending memorised associations, textual recognition, learned chemical regularities and genuine inferential behaviour in ways that ordinary benchmark scores cannot disentangle.

The lesson is consequently not that molecular AI has failed.

It is that **our examination of molecular AI has become too easy**.

The next generation of scientific benchmarks must distinguish recollection from extrapolation, representation matching from molecular understanding, and retrospective familiarity from prospective discovery.

Ultimately there is one wonderfully simple experiment.

Give the machine a molecule whose relevant property **has never been measured**.

Demand its prediction.

Seal the answer.

Then enter the laboratory and measure reality.

If experiment agrees, there can be no molecular déjà vu—because there was nothing to remember.

At that moment the machine will no longer merely have recognised chemistry's past.

It will have successfully anticipated chemistry's future.

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